ORTHOPEDIC NURSING

BY JANE TERER

EXPECTATIONS

COURSE OBJECTIVES

MAIN OBJECTIVE

By the end of this Course, the learner will gain the necessary knowledge, skills and attitudes to promote health, prevent illness, diagnose, co-ordinate, manage and rehabilitate infants, children and adults suffering from orthopedic conditions.

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SPECIFIC OBJECTIVES

1. Revise the structure and functions of the different tissues that compose the musculoskeletal system.
2. Outline bone healing
3. Describe the components of the nursing assessment, formulate a nursing diagnosis and give appropriate care to the person with musculoskeletal conditions; including: congenital talipes, osteogenic imperfect, hip dysplasia, rheumatoid arthritis, osteomylitis, TB of spine, scoliosis, ankylosis, traumatic fractures (potts, colles, humerus, radial/ulna pelvis, hip, femoral shaft/neck, tibia/fibula, spine, bone neoplasm's, degenerative osteoporosis)

Specific Objectives cont'd

3. Explain the diagnostic test indicated for the person with musculoskeletal problem, the rationale for each test and appropriate nursing responsibilities associated with each test.

4. Describe the role of the nurse in the management of fractures and prevention of fracture-related complications.

Supporting Objectives

1. Review basic anatomy and physiology of bones and joints related to Orthopaedic nursing
2. Describe fractures and soft tissue injuries
3. Describe Orthopaedic inflammatory conditions:
   - Osteomyelitis
   - Rheumatoid Arthritis
   - Osteoarthritis
   - Pyogenic arthritis
   - Gout
   - Osteoporosis

Supporting Objectives Cont'd

5. Describe musculoskeletal system tumors
6. Describe congenital abnormalities of the musculoskeletal system (Talipes and Congenital hip dislocation/hip dysplsia), osteogenic imperfect
7. Care of the patient undergoing amputation

COURSE CONTENT

- Review of Anatomy and Physiology of the Musculoskeletal System, bone healing
- Fractures and Soft Tissue Injuries
- Orthopaedic Inflammatory Conditions
  - Osteomyelitis
  - Pyogenic Arthritis
  - Rheumatoid arthritis and Osteoarthritis
  - Gout
  - Osteoporosis

Course Content cont'd

- Tumors of the Musculoskeletal system
- Congenital abnormalities of the musculoskeletal system
  - Talipes equinovarus
  - Congenital hip dislocation/hip dysplasia
  - Osteogenic imperfect
- Diagnostic tests and Investigations indicated in Musculoskeletal pathologies
- Medical management and the Nursing care of individuals with Musculoskeletal disorders.

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INTRODUCTION

Orthopedics is the branch of medicine that deals with disorders or deformities of the skeletal system and associated muscles, joints and ligaments.

Orthopedic Nursing is a specialty focused on prevention and treatment of musculoskeletal disorders using the appropriate and scientific based nursing care.

ANATOMY AND PHYSIOLOGY OVERVIEW

ANATOMY AND PHYSIOLOGY OVERVIEW

The Musculoskeletal system function is interdependent with other body systems

The bony skeleton provides a supportive framework for body structures.

Bones also stores Ca^{2+} (98%), Phosphorous, magnesium and fluoride ions.

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Anatomy and Physiology Overview cont'd

- Red bone marrow produce red and white blood cells (*Hematopoiesis*)
- Joints hold bones together and allow movement.
- Muscles attached to bones helps moves bones and aid in heat production (maintain body temperature)

Anat… and Physio… cont'd

Bone Cell Types:

(i) ***Osteoblasts*** for bone formation

(ii) ***Osteocytes*** for bone maintenance

(iii) ***Osteoclasts*** for destroying, resorbing and remodeling bone substance

Other related concepts

Ligaments

Ligaments are parallel bands of flexible, dense fibrous connective tissue whose primary function is to connect the articular ends of bones and provide stability.

They permit movement in some directions but limit movement in other directions hence preventing joint injury, as is the case with knee and hip joints.

Ligaments also attach to soft tissue to suspend structures e.g. the Suspensory ligament of the ovaries.

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Tendons

Tendons are bands of dense fibrous tissues forming the origin and insertion of muscles.

They are aligned with sheaths, which is again lined with synovial membrane that provide lubrication for each tendon movement.

The sheaths enclose certain tendons especially in the wrist and ankle.

Fascia

This is a sheet of loose connective tissue that may be found directly under the skin as ***superficial fascia*** or as a sheet of dense fibrous connective tissue making up a sheath of muscles, nerves and blood supply.

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Bursae

Are small sacs of connective tissue located whenever pressure is exerted over moving parts, hence preventing injury to muscle tendons.

Bursae are lined with synovial membranes and contain synovial fluid, which serve as cushion between the moving parts.

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Joints

Are of three major types:

(i) *Fibrous (Synarthroses) joints:* Allows no movement at all e.g. the sutures of the Skull.

(ii) *Amphiarthroses (Cartilaginous) joints:* Allows little movement e.g. the intervertebral joints and the joints at the pubic symphysis.

(iii) *Diarthroses (Synovial):* Allows free movements e.g. the hip joints.

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Factors affecting bone formation

1. Factors influencing/responsible for bone formation:
   - Calcium level
   - Phosphorus
   - Enzyme Alkaline phosphatase (ALP)
   - Calcitonin
   - Vit. D
   - PTH

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Factors enhancing bone formation cont'd

- Growth Hormone (GH)
- Glucocorticoids (Cortisol)
- Sex hormone (Estrogen, Androgens)

Factors limiting bone formation

- Infection and inflammation
- Activity and Weight bearing

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PHYSIOLOGY OF BONE HEALING

The process of bone healing is known as ***Callus Formation.*** Fractures and surgical interruptions of bone both heal by the same process and occurs in five major stages as follows:

- Hematoma formation
- Fibrin meshwork formation
- Invasion of oesteoblast
- Callus formation
- Remodeling

1. Hematoma formation

Because bone is highly vascular, bleeding occurs at both ends of the fractured bone. Increased capillary permeability permits further extravasation of blood into the injured area.

Blood collects in periosteal sheath or adjacent tissues and fastens the broken

2. *Fibrin meshwork formation*

Fibroblast invade the hematoma forming a fibrin meshwork.

White blood cells wall off the area localizing the inflammation.

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3.Invasion of oesteoblast

As oesteoblasts invade the fibrous union to make it firm, blood vessels develop from capillary buds, thereby establishing a supply for nutrients to build collagen and granulation tissue is formed.

Collagen strands become longer and begin to incorporate Ca^{2+} deposits leading to formation of cartilage.

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4. Callus formation

Osteoblasts form a woven bony structure known as ***callus***.

The osteoblasts continuously lay a network for the build up of bone, while osteoclasts destroy dead bone and help in the synthesis of new ones.

Ca^{2+} and phosphorus are deposited as mineral salts.

5. Remodeling

Excess callus is reabsorbed and new bone is laid down.

This is important because bone that has not undergone remodeling lacks the mechanical properties necessary for weight bearing.

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Factors impeding callus formation

1. Inadequate/poor reduction of the fracture

2. Excessive edema at the fracture site impeding the supply of nutrients to the area of injury.

3. Excessive bone loss at the time of injury, which prevents sufficient bridging of the broken ends.

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Factors Impeding Callus formation Cont'd

4. Inefficient immobilization
5. Infection at the site of injury.
6. Bone necrosis.
7. Anemia or other systemic conditions.
8. Endocrine imbalance.
9. Poor nutritional status.

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FRACTURES

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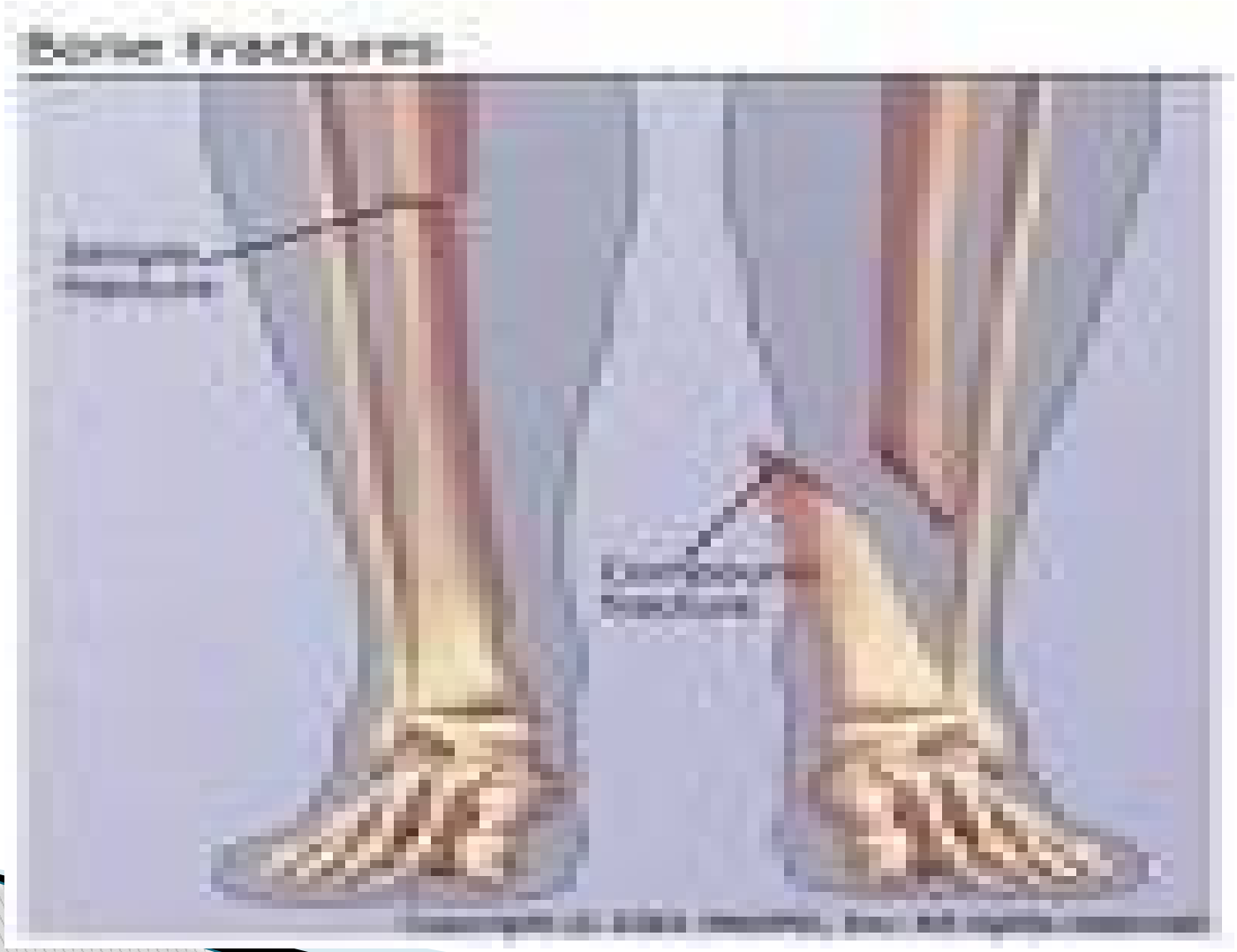

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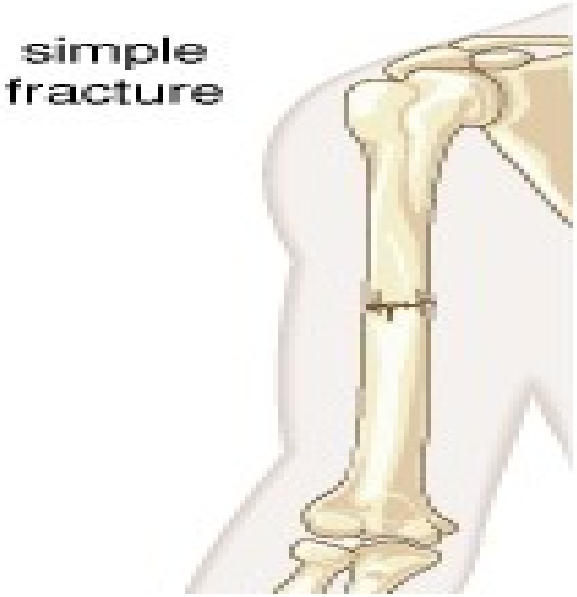

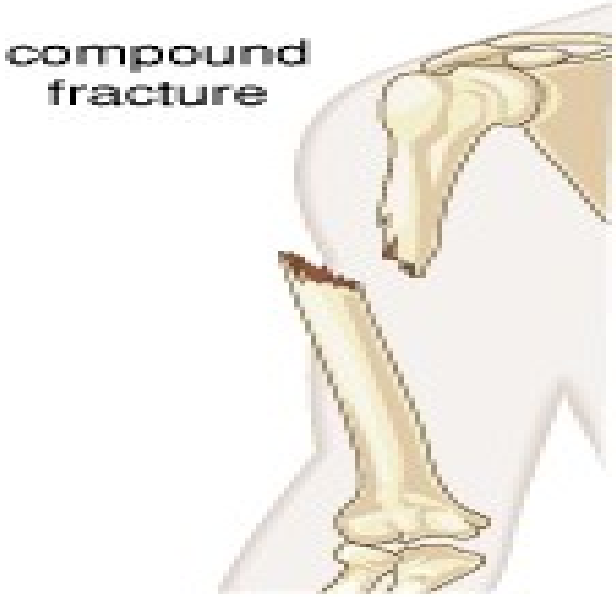

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FRACTURES

A fracture is a discontinuity or break in a bone related to a blow, fall, accident, stress or disease process.

Avulsion fractures occur when a strong ligament or tendon pulls a fragment of bone away from the rest of bone.

Broken bones affect other adjacent structures.

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Effects of fractures

- Soft tissue edema
- Hemorrhage into muscles and joints
- Joint dislocation
- Ruptured tendons
- Severed nerve
- Damaged blood vessels
- Damaged body organs

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Classification/Types of fractures

Fractures can either be:

1) *Complete:* There is a break across the entire bone;

2) *Incomplete:* Breakage occur partially;

3) *Closed fracture:* There is breakage but the skin remains intact; *OR,*

4) *Open fracture:* Involves the skin or mucus membrane.

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Complete fractures

Complete fractures can be/includes

1) Simple(Closed)
2) Compound(Open)
3) Comminuted (Involves more than two fracture fragments)
4) Linear
5) Oblique
6) Spiral
7) Transverse

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Complete fracture cont'd

8. Impacted
9. Pathological
10. Avulsion
11. Extracapsular
12. Intracapsular

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Incomplete fractures

Incomplete fractures include

(i) Greenstick fracture

- *Torus fracture:* There is buckling of the cortex caused by impaction;
- *Bowing fracture:* The bone becomes curved along its longitudinal axis; *OR*
- *A transverse fracture occurring* in the cortex and extending into the midportion of the bone but then becomes oriented along the longitudinal axis of the bone without disrupting the opposite cortex.

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Incomplete fractures cont'd

(ii) Stress fracture: A very small crack in bone caused by unusual or repeated stress or heavy continuous weight on the ankle or leg; typically occurs in weight-bearing bones, such as the tibia and metatarsals bones and is a common sports injury, mostly associated with athletics.

(iii) Transchondral fracture: Involves entire thickness of cartilage down to subchondral bone.

Grading of Open fractures

Open fractures can be graded according to level/degree of contamination as follows:

(i) ***Grade 1:*** Fracture with clean wound ≈1 cm or less.

(ii) ***Grade 2:*** Has a large wound with extensive soft tissue damage.

(iii) ***Grade 3:*** The wound is contaminated and there is extensive soft tissue damage.

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CLINICAL FEATURES OF FRACTURES

Common clinical manifestations of fractures include:

(i) Pain (swelling, muscle spasm)

(ii) Loss of normal function

(iii) Obvious deformity

(iv) Excessive motion at the site

(v) Crepitus – *the rattling sound or vibration produced by rubbing bone or irregular cartilage surfaces together.*

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Clinical features of fractures cont'd

(vi) Soft tissue edema

(vii) Warmth over injured area

(viii) Ecchymosis

(ix) Impairment or loss of sensation/paralysis

(x) Signs of shock

(xi) Evidence of fracture on X-ray film

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DIAGNOSIS OF FRACTURES

Read and make notes on:

1. ***Laboratory and Radiological tests in the Medical Diagnosis of Fractures.***

2. ***History and Physical examination of individuals with musculoskeletal disorders.***

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MANAGEMENT OF FRACTURES

Objectives of Fracture Management

The major focus during the management of fractures aims at;

1. Reducing the fracture,
2. Immobilization, and
3. Rehabilitation.

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IMMEDIATE MANAGEMENT

1. Maintain airway and assess for shock
2. Splint the fracture
3. Preserve correct body alignment

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SECONDARY MANAGEMENT

Simple Fracture:

1. Reduction: *closed reduction/manual manipulation*
2. Traction
3. Immobilization (cast, wires, screws, plates)

Compound fracture

1. Surgical debridement
2. Wound culture
3. Wound dressing
4. Monitor infection (osteomyelitis, tetanus, gas gangrene)
5. Wound closure
6. Reduce fracture
7. Immobilize the fracture

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Reduction

Is the restoration of the fracture fragments to anatomic alignment and rotation.

Reduction can be achieved in two ways:

1. *Closed reduction:* Aligning the bone through external manipulation/traction (cast, splint or other devices).

2. *Open reduction*: A surgical procedure (ORIF-Open reduction with internal fixation)

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Disadvantages of internal fixation

(i) Hypersensitivity to the devices; some patients may react to the internal fixator devices.

(ii) Damage to the bone and underlying tissues during the procedure.

(iii) Infections.

(iv) Bone overgrowth may occur as the bone excludes this plate.

(v) High chances of failure rate.

(vi) It is expensive as it may need to be removed later.

Immobilization

Immobilization is used to hold broken bones in contact to each other until healing takes place.

This can be achieved externally through external fixators such as casts, splints, braces and traction; or internally by use of metal plates and pins.

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Traction

Is a mechanism that ensures a study pull is exerted on a part of or parts of the body to maintain adequate alignment.

Traction is used to reduce and immobilize fractures, and to maintain correct alignments.

It also helps in overcoming muscle spasms, correcting deformities and stretching adhesion.

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Types of traction

1. **Skin traction**

   The traction device is applied directly to the skin and attaching weights to them (Buck's extension and Russel traction)

   Hold 2-3.5 Kg (4.5 to 8 lb)

   Pelvic traction-4.5-to 9 Kg (10 to 20 lb)

   Monitor skin break down as a complication of traction

Skin traction

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Skin traction

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Skeletal traction

The traction device is applied directly to the bone.

Can hold 7-12 Kg (15-25 lb).

Potential complication include infection

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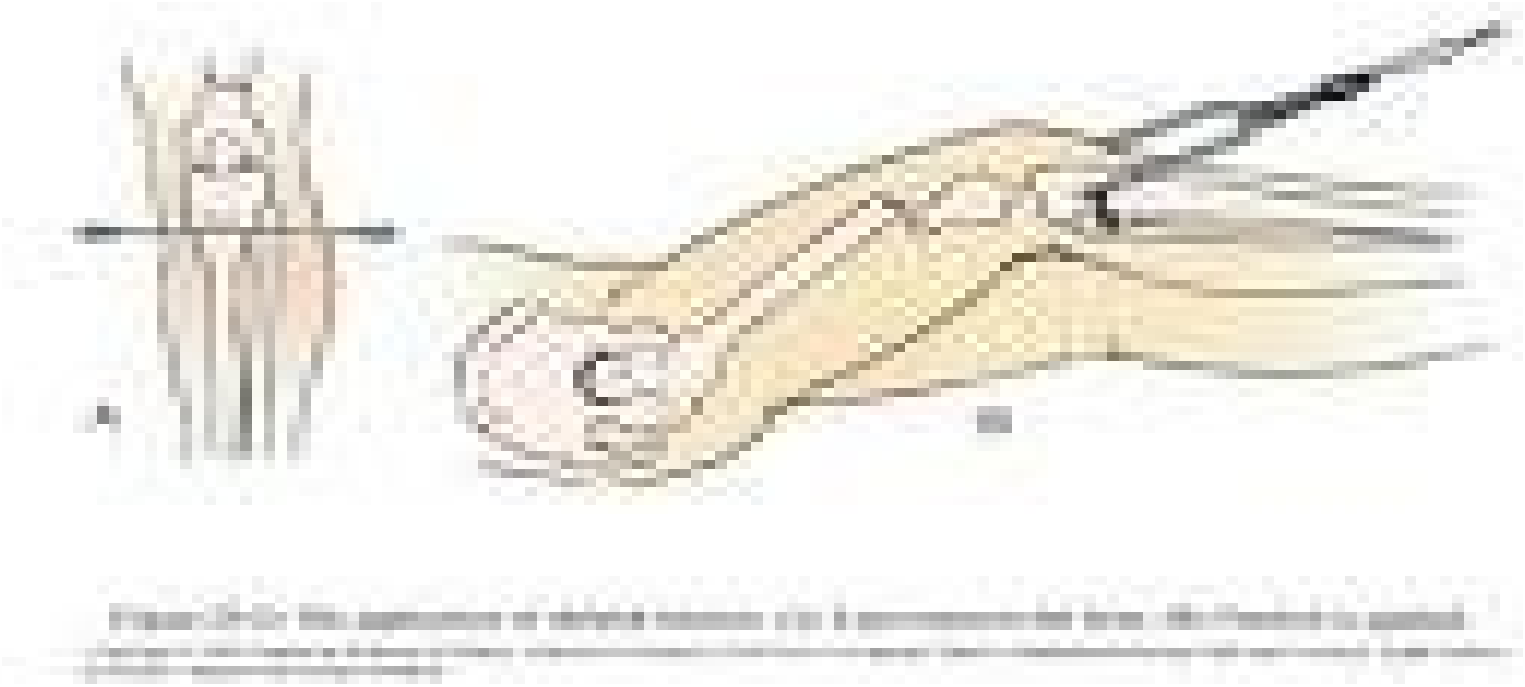
Skeletal traction



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Principles of effective traction

1. Use counter-traction to achieve effective traction.
2. Traction must be continuous.
3. Skeletal traction is never interrupted.
4. Do not remove weight until advised

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5. Always position the patient at the centre of the bed to help avoid poor body alignment.

6. Ropes must be unobstructed

7. Weight must hang free and not rest on the floor.

8. Knots on the rope or footplate must not touch the pulley or the foot of the bed.

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CAST

Is a rigid external immobilizing device that is moulded to the contours of the body

Major purpose of the cast include:

(i) To immobilize an already reduced fracture.

(ii) To correct the deformity.

(iii) To stabilize weakened joints.

Casting Materials

- ***Fiberglass casts***: Composed of water-activated polyurethane materials with the versatility of plaster but are more water resistant, lighter in weight and more durable than plaster

- ***Plaster Casts:***
  Consists of rolls of plaster of Paris impregnated bandages wet in cool water and smoothly applied to the body.

  Produces a rigid dressing within **15 – 20** minutes, but require a **1 – 3** days to dry completely depending on plaster thickness and environmental drying conditions.

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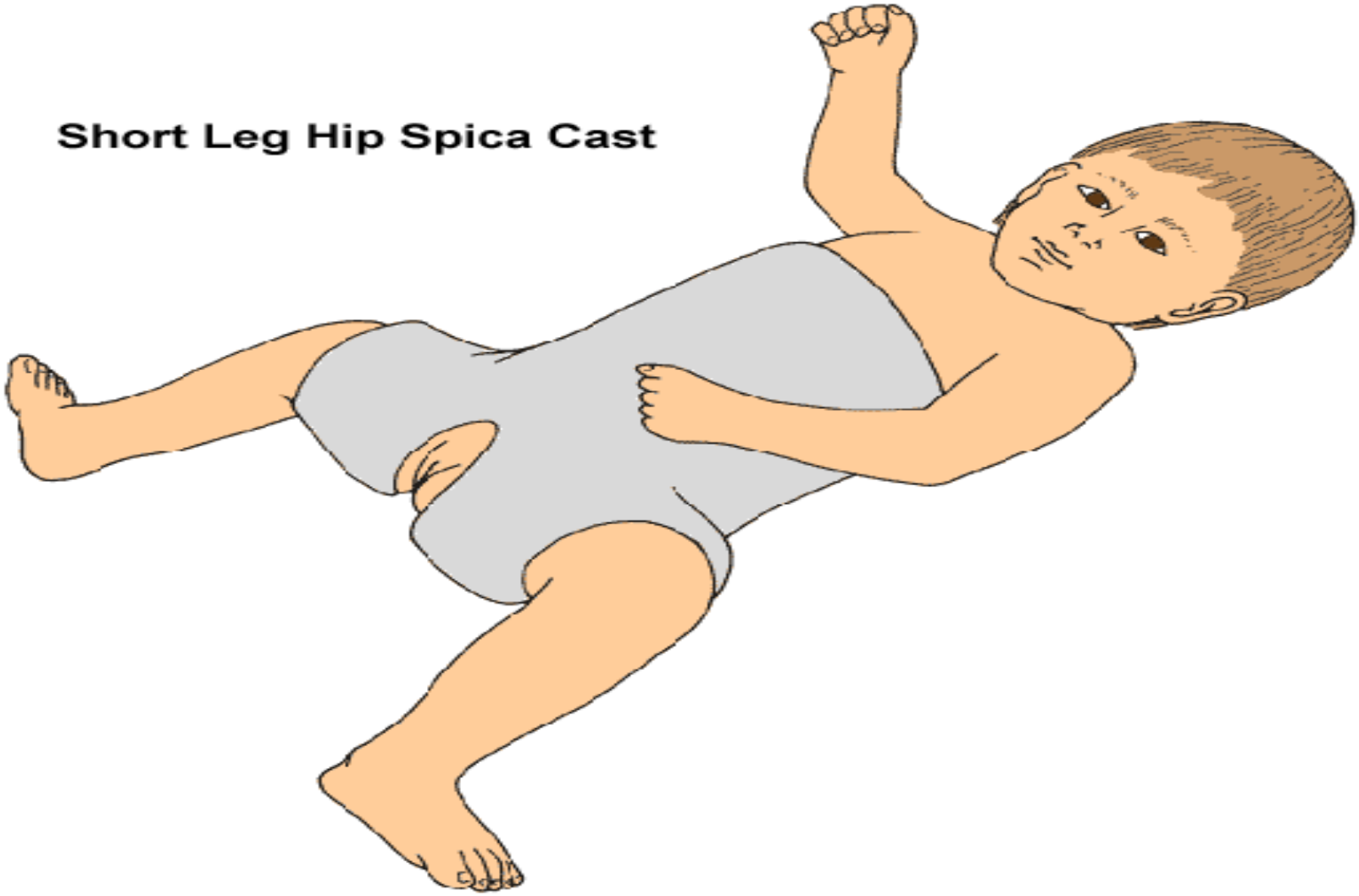
Types of cast



Types of casts

1. ***Short arm cast:*** Extends from below the elbow joint to the palmar crease, secured around the base of the thumb.

2. ***Long arm cast:*** Extends from the axillar fold to the proximal palmar crease, with the elbow immobilized at right angles

3. ***Short leg cast:*** Extends from below the knee to the base of the toes with the foot flexed at right angles in a neutral position.

Types of Casts cont'd

1. ***Long leg* cast:** Extends from the junction of the upper and middle 1/3rd of the thigh to the base of the toes; the knees may be slightly flexed.

2. ***Walking cast***: A short or long- leg cast reinforced for strength

3. ***Body cast:*** Encircles trunk

4. ***Shoulder/hip spica***: Body jacket that encloses the trunk shoulder and elbow for shoulder spica while the hip spica encloses the trunk and lower extremities.

Potential complications of a cast

(i) Compartment Syndrome

(ii) Pressure ulcer

(iii) Disuse syndrome

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Care of the Patient on a cast

Open fractures may not be treated with casts initially or if a cast is to be applied, a window may be required for dressing change

(i) The plaster must be kept dry

(ii) Assess skin under the cast for skin integrity

(iii) Spica casts should be avoided in abdominal distension

Assessment of pt on cast

Prevent neurovascular complications by assessing the 5 P's:

(i) Pain
(ii) Parasthesia
(iii) Pulses
(iv) Palor (colour)
(v) Paralysis

Together with edema, temperature and capillary refill.

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Bad/Danger signs may include:

(i) Progressive unrelieved pain

(ii) Parasthesia

(iii) Motor loss

(iv) Sensory loss

(v) Sensation of tightness

(vi) Coolness

(vii) Paleness

(viii) Slow capillary refill (<2-4 seconds)

Preoperative care of patient undergoing orthopedic surgery

1. Assess and for hydration status, medication histories and possible infections

2. Relieve pain through physical, psychological and pharmacologic strategies

3. Maintain adequate neurovascular function through assessment and prompt intervention

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Pre-operative cont'd

4. Promote health by teaching coughing techniques, deep breathing and monitor fluid intake.

5. Improve mobility by elevating extremities and controlling pain.

6. Help pt maintain self esteem.

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Post-operative care

1. Relieve pain.
2. Maintain adequate neurovascular function.
3. Improve and maintain physical mobility.
4. Promote and manage potential complications.
5. Promote home and community based care

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Compartment syndrome

A complication of trauma in which there is increased pressure within a limited anatomical space compromising circulation, viability and function of the tissues within that space.

Restrictive space may be due to dressing, splint or even a cast.

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Compartment syndrome

Compartment syndrome

Pathophysiology

Trauma------ fluid accumulates in compartment ---------- increased pressures --------- decreased blood supply---------- tissue hypoxia ------- increased permeability --------- compartment pressure continue to rise.

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Management of C. Syndrome

Goals of management include:

1. To decrease tissue pressure.

2. To restore blood flow.

3. To preserve function of the affected limb.

Removal of external compression device by splitting cast may decrease the pressure

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Management cont.

Surgical intervention;

- Decompressive fasciotomy is indicated if conservative management fails.

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Total Hip Replacement

Replacement of severely damaged hip with artificial joint.

Indicated mostly in patients over 60 years of age where regenerative bone healing may not be achievable following:

(i) Arthritis (degenerative joint disease, Rheumatoid arthritis).
(ii) Femoral neck fracture.
(iii) Failed prosthesis/osteotomy
(iv) Congenital hip disease

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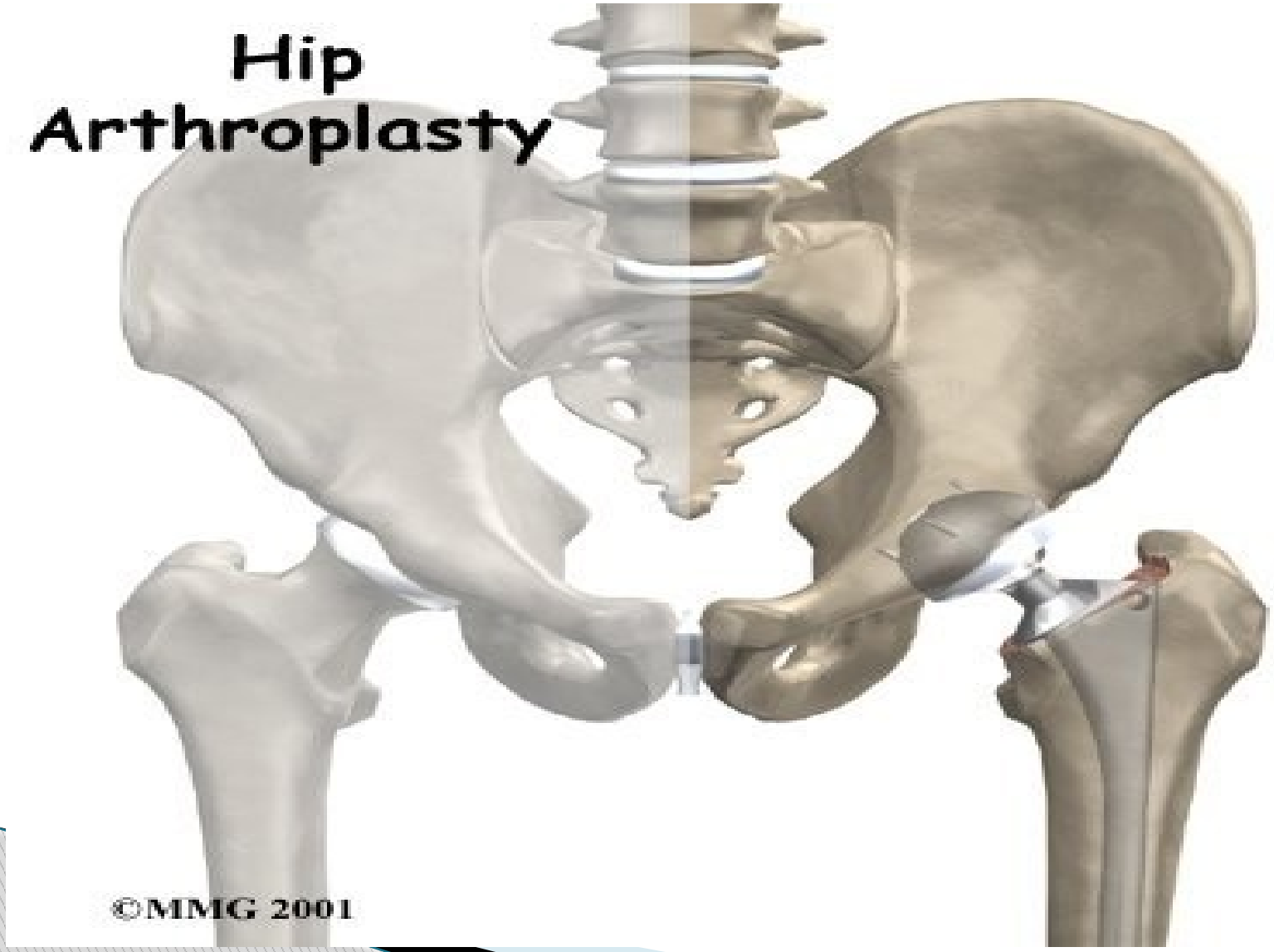
Hip Replacement cont'd



Hip Replacement cont'd

NURSING CARE OF THE PATIENT WHO HAS UNDERGONE TOTAL HIP REPLACEMENT

Nursing Care Cont'd

1. Monitor for possible complications such as:

(i) Hip prosthesis dislocation

(ii) Excessive wound drainage.

(iii) Thromboembolism.

(iv) Infection.

(v) Heel pressure and possible pressure sores

Nursing care cont'd

2. Monitor for other complications such as:

(i) Heterotrophic ossification.

(ii) Vascular necrosis.

(iii) Loosening of the prosthesis.

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Nursing care cont'd

3. Prevent hip dislocation by;

(i) Maintaining the femoral head in the acetabulum.

(ii) Nursing the affected leg in a slightly abducted position.

(iii) Using an abduction splint or 2 – 3 pillows placed between the legs (wedge pillow).

Nursing care cont'd

4. Pt's hip is never flexed more than 60 degrees

5. Flex the unaffected hip and use trapeze when giving bed pan.

6. Teach the patient not to flex the affected hip.

7. Use an abduction splint when moving the patient out of bed.

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Signs of a possible dislocation include

(i) Increased pain at surgical site, swelling and immobilization.

(ii) Acute groin pain at the affected hip or increased discomfort.

(iii) Shortening of the affected leg.

(iv) Abnormal extension or internal rotation.

(v) Restricted ability/inability to move.

(vi) Report of a “popping” sensation by the patient

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SOFT TISSUE INJURIES

Soft Tissue Injuries

Contusions

Strains

Sprains

Contusions

A Contusion is a soft tissue injury produced by a blunt force such as a blow, a kick or even a fall.

The small blood vessels rupture and bleed into soft tissues leading to ecchymosis or a bruise.

Hematoma may develop if bleeding is pronounced.

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Symptoms of Contusion include

(i) Pain
(ii) Swelling
(iii) Joint dislocation

Management strategies include

- Intermittent application of cold compresses
- Resting the affected limb

Most contusion resolve within 1-2 weeks

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Strains

A strain is a muscle pull caused by overuse, overstretching or excessive stress on a muscle.

It is a microscopic, incomplete muscle tear, with some bleeding into the tissue.

Signs may include

(i) Isometric muscle contraction

(ii) Sudden pain with local tenderness on muscle.

Sprains

A sprain refers to an injury to the ligaments surrounding a joint, commonly caused by a wrench or twisting.

A torn ligament loses its stabilizing ability

Common symptoms of sprain include:

- Oedema
- Tenderness which increase within 2-3 hours
- Decreased movement

An X-ray film should be taken to rule out an avulsion fracture

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Management of Soft tissue Injuries

The management of contusions, strains and sprains can be summed up with the acronym "***RICE***"

R = Rest

Resting the affected limb prevent additional injury and promote healing.

I = Ice

Application of cold compresses produces vasoconstriction hence decreasing the edema and discomfort associated with such injuries.

C = Compression

An elastic bandage provides pressure that controls bleeding, reduces edema and also supports the underlying structures.

E = Elevation

Elevating a limb controls swelling by increasing venous return to the central circulation.

Surgical repair is done for tone torn muscle and disrupted ligament.

JOINT DISLOCATION

Joint Dislocation cont'd

Is a condition in which the articular surfaces of the bones forming a joint are no longer in anatomical approximation to each other causing pain and impaired movement at such joints.

The impairment of the approximation of bone surfaces can be total or incomplete/partial, in which case the condition is referred to as a ***subluxation.***

Joint dislocation cont'd

Dislocations can also be classified as:

- ***Congenital:*** present at birth
- ***Spontaneous*** or
- ***Traumatic***

A traumatic joint dislocation constitutes an ***orthopedic emergency*** which can lead to avascular necrosis and nerve palsy if not promptly treated, because the associated joint structures are distorted and severely stretched.

Joint dislocation commonly manifests with:

(i) Pain on the affected joint and around it.
(ii) Change in joint contour
(iii) Change in length of the extremity involvd
(iv) Loss of normal mobility
(v) Change in the axis of the dislocated bones

An X-ray film can be taken to help confirms the diagnosis

Management of J. Dislocation

1. Immobilize the affected joint/s
2. Promptly reduce the joint
3. Control pain
4. Monitor neurovascular status
5. Perform range of motion(ROM)
6. Educate pt on how to immobilize joint

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ORTHOPEDIC INFLAMMATORY CONDITIONS

Common Inflammatory Conditions of Bones and Joints

Osteomyelitis

Arthritis:

- Rheumatoid arthritis
- Septic arthritis
- Osteoarthritis

Gout

OSTEOMYELITIS

Osteomyelitis

Refers to infection of bone, which can occur following:

- Extension of soft tissue infection.
- Direct bone contamination.
- Hematologic(blood-borne)dissemination

Osteomyelitis

Predisposing factor to Osteomyelitis

(i) Malnutrition

(ii) Elderly /Old age

(iii) Obesity

(iv) Impaired immune systems

(v) Post operative surgical wound which occur within 30 days after surgery

Causative organisms of Osteomyeliti

Staphylococcus aureus is responsible for approximately 70 – 80% of all bone infections.

Other causative organisms of bone infections include:

- *Proteus spp.*
- *Pseudomonas spp.*
- *Escherichia coli*

Osteomyelitis cont'd

Bone infections are more difficult to eradicate than soft tissue infections.

This is because the affected bones become walled off; natural body immune responses are blocked and there is less penetration by antibiotics

Osteomyelitis may become chronic and affect quality of life.

Pathophysiology of Osteomyelitis

The initial response to infection is inflammation, increased vascularity, and edema. After 2 or 3 days, thrombosis of the blood vessels occurs in the area, resulting in ischemia with bone necrosis.

The infection extends into the medullary cavity and under the periosteum and may spread into adjacent soft tissues and joints.

Pathophysiology cont'd

Unless the infective process is treated promptly, a bone abscess forms.

This abscess cavity contains dead bone tissue called a ***sequestrum,*** which does not easily liquefy and drain.

Therefore, the cavity cannot collapse and heal, as occurs in soft tissue abscesses. New bone growth, called ***involucrum,*** forms and surrounds the sequestrum.

Pathophysiology cont'd

Although healing appears to take place, a chronically infected sequestrum remains and produces recurring abscesses throughout the patient's life causing a chronic osteomyelitis.

Clinical features of Osteomyelitis

(i) Onset is sudden

(ii) High fever and chills

(iii) Rapid pulse

(iv) General body malaise

(v) Pain on the infected area.

Clinical features cont'd

(vi) Extreme tenderness

(vii) Chronic draining sinus.

(viii) Recurrent period of chronic pain.

(ix) Chronic inflammation and swelling

Diagnosis

- Signs and symptoms
- MRI
- Elevated leukocyte level
- Increased sedimentary rate
- Wound culture
- X-ray (early finding demonstrate soft tissue swelling)

Management of Osteomyelitis

Prevention of osteomyelitis should be the goal.

- No elective orthopeadic surgery with infection.
- Observe sepsis during surgery.
- Prompt management of soft tissue infections.
- Prophylactic antibiotics before invasive procedure.

Management cont'd

During treatment the goal is to control infection

1. Administer antibiotics
2. Ensure hydration
3. Provide appropriate diet
4. Correct anaemia
5. Immobilize the affected area
6. Apply warm wet soaks
7. Surgical-(sequestrectomy and debridement)

RHEUMATOID ARTHRITIS

Rheumatoid Arthritis

Arthritis is an inflammation of the joint, causing pain and stiffness.

Rheumatoid arthritis is a chronic, systemic disease characterized by recurrent inflammation of the diarthrodial joints, related structures and the surrounding tissues.

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Aetiology

The exact cause of RA is not clearly understood

Theories suggesting:

(i) ***Infection*** e.g with Epstern-Barr virus, the parvoviruses and mycobacteria may trigger the process.

(ii) ***Auto-immunity***

(iii) ***Genetic factors***

(iv) ***Others:*** such as metabolic and biochemical abnormalities, nutritional, environmental and occupational factors.

Pathophysiology of R. Arthritis

The disease progresses through four stages:

1. The unknown aetiologic factor initiate synovitis with swelling of the synovial membrane producing excess synovial fluid.

2. Inflammatory granular tissue called ***pannus*** is formed at the junction of the synovial membrane and cartilage.

   This eventually spreads and invades the joint capsule and subchondrial bone.

Pathophysiology cont'd

3. A tough fibrous connective tissue replaces the pannus thus occluding the joint space. This results into ***fibrous ankylosis*** causing a decrease in joint motion and increased deformity.

4. As the fibrous tissue calcifies, ***bony ankylosis*** may result in total joint immobilization.

Systemic manifestations of R. Arthritis

- Pulmonary
- Cardiac
- Vascular
- Ophthalmological
- Dermatological
- Hematological

Clinical features of R. Arthritis

The signs and symptoms of RA are quite ***non specific and insidious.*** These may include

1. Fatigue
2. Anorexia
3. Weight loss, fever, malaise, morning stiffness of the joints.
4. Pain during rest and movement; night pains, edematous, erythematous “boggy joints”
5. History of precipitating stressful event

Clinical features of RA cont'd

6. Limitation of motion.
7. Signs of inflammation.
8. Bilateral joint symptoms (small joints).
9. Cervical joints may be affected.

Advanced stage of R. Arthritis

Later symptoms

- Pallor
- Anemia
- Colour changes of digit
- Muscle weakness
- Joint deformity

Later symptoms of RA cont'd

- Paraesthesias
- Decreased joint mobility
- Contractures
- Subluxation
- Dislocation
- Increased pain

Diagnosis

- History and physical exam
- Positive Rheumatic factor
- Biopsy
- Presence of immune complex and WBC in synovial fluid

Management of Rheumatoid Arthritis

The goal of therapy is to:

1. Relieve symptoms

2. Prevent joint destruction

3. Maintain joint functions

4. Promote independence and quality of life

Management of RA cont'd

- Administer drug to control pain (NSAIDS, Corticosteroids)
- Stabilize /Support the joints
- Protect a joint or body part from external trauma.
- Assist the patient to exercise specific joints
- Apply cold pacts
- Rest, exercise and diet are important

Management of RA cont'd

- Surgery may be indicated to correct the deformity, relieve pain and restore function.

- Educate the patient on balance of rest and activity, joint protection and energy conservation, proper use of medication and safety measures to prevent injury

SEPTIC ARTHRITIS

Septic Arthritis

In S. Arthritis, the joint become is either infected through infections from another part of the body or directly through trauma or surgical instrumentation.

Factors that may predispose to S. Arthritis include:

(i) Trauma to the joints.
(ii) Joint replacement
(iii) Coexisting arthritis
(iv) Diminished host resistance

Causative organisms of septic arthritis

The leading causative agent is *Staphylococcus aureus,* followed by *Streptococci* and some gram negative organisms have also

Prompt recognition and treatment of infected joints are important because accumulating pus results in ***chondrolysis*** (destruction of hyaline cartilage)

Clinical Manifestations of S. Arthritis

The patient with acute septic arthritis usually presents with:

(i) A warm, painful, swollen joint with decreased range of motion.

(ii) Chills and fever

(iii) Leukocytosis.

Diagnostic measures for S. Arthritis

(i) Signs and symptoms

(ii) Culture of synovial fluid

(iii) Computed tomography

(iv) MRI

Management strategies of S. Arthritis

(i) Broad-spectrum IV antibiotics are started promptly and then changed to organism-specific antibiotics after culture results are available.

(ii) Needle aspiration ***(Arthrocentesis)*** to remove excess fluid exudate, and debris.

(iii) Immobilize the joint in a functional position

S. Arthritis Management cont'd

(iv) Administer analgesia and anti-inflammatory agents such as NSAIDS.

(v) Ensure proper/adequate nutrition and hydration.

(vi) Educate the patient on medication, the septic arthritis process, importance of supporting the affected joint and strategies to promote healing through aseptic dressing changes and proper wound care.

Osteoarthritis

Also known as Degenerative Joint Disease (DJD), this is a slowly progressive disorder of articulating joints, especially in the weight bearing joints.

It is characterized by degeneration of articular cartilage and overgrowth of bone.

Damage is localized to the joints and surrounding tissues

Aetiology of OA

May occur as ***Primary (Idiopathic) Osteoarthritis*** or ***Secondary***, resulting from a previous joint injury or inflammatory disease.

Risk factors to Secondary OA may include:

(i) Previous fractures.

(ii) Infections

(iii) Congenital deformities

(iv) Old age - usually seen in 50-70 years but may also occur as early as 45 years

Aetiology of OA cont'd

Other factors influencing development of osteoarthritis include:

- *Osteochondritis of head of femur* esp. in children (congenital structural defects).
- *Metabolic disturbances* such as obesity and overweight.
- *Repeated intra-articular hemorrhage*
- *Septic arthritis*
- *Excessive stress on joints* through occupational or recreational repeated use.

Pathophysiology of OA

The factors responsible for OA development (genetic and hormonal factors, mechanical injury, previous joint damage or any other factor) stimulates a ***Chondrocytic response*** in the affected joint.

The stimulated chondrocytes ***releases chemokines and cytokines.***

This leads to the ***Stimulation, production, and release of proteolytic enzymes***, metalloproteases, collagenases.

Pathophysiology of OA cont'd

The resulting damage caused by the *proteolytic enzymes and* metalloproteases, further predisposes to ***more chondrocyte response*** leading to further damage.

With degeneration over time, ***bones become yellow and opaque with rough surfaces*** and areas of malacia (softening).

New bone outgrowth are formed at the joint margins and at the attachment sites of ligament.

Clinical manifestation of OA

(i) Joint enlargement and Crepitus

(ii) Pain which increases with weight bearing

(iii) Limited joint motion

(iv) Morning stiffness (less than one hour)

(v) Non-inflammatory effusion

(vi) Knocked knees

Management of OA

There is no specific management but therapy is aimed at symptomatic relief and control of pain.

Encourage appropriate nutritional intake to maintain ideal weight.

Behavioural change especially with occupation and exercises.

Surgical Intervention may be indicated.

GOUT

Gout is a heterogeneous group of conditions related to a ***defect of purine metabolism*** resulting in ***hyperuricemia*** (a uric acid serum concentration > 7 mg/dL [0.4 *f*mol/L]).

Oversecretion of uric acid or a **renal defect** resulting in decreased excretion of uric acid, or a combination of both, occurs.

The incidence increases with age and body mass index and it tends to occur more commonly in males than females.

Gout cont'd

Is the principal manifestation of sustained hyperuricaemia.

Men are ten times more likely to have gout than women (At puberty male uric level rise and remain higher than those of females until menopause when female uric level rise)

Serum uric acid are related to Age, sex and genetic constitution.

Gout Classification

Gout can be classified as **Primary** or **Secondary.**

In ***Primary hyperuricemia***, elevated serum urate levels or manifestations of urate deposition are due to faulty uric acid metabolism.

Primary hyperuricemia may be due to severe dieting or starvation, excessive intake of foods that are high in purines (shellfish, organ meats), or heredity.

Classification of Gout cont'd

In ***Secondary hyperuricemia***, gout is a clinical feature secondary to any of a number of genetic or acquired processes.

Such processes may include conditions in which there is an increase in cell turnover such as leukemia, multiple myeloma, some types of anemias, psoriasis and even an increase in cell breakdown.

Gout cont'd

About 60% of uric acid is replaced daily and approximately 75% of uric acid is excreted in the kidney and remainder lost in the gut.

Uric acid is filtered at glomerulus and 90% is then reabsorbed

Causes of hyperuricemia

- Prolonged use of diuretic drugs are the main cause but clinical gout is unknown
- Renal failure
- Increased production of uric acid: polycthaemia, hemolytic anemia, severe psoriasis and carcinomatosis.
- Decreased renal excretion of uric acid-chronic renal disease, drug administration, reduction in fractional urate clearance

Causes of hyperuriceamia cont'd

The patient with Gout is usually obese, and drinks more alcohol. Dietary purines have a modest effect on plasma uric level

Short term alteration in diet such as starvation and ingestion of large amount of alcohol may cause hyperuricaemia by reducing renal clearance of uric acid due to increased lactic acid production which competes with uric acid excretion in the kidneys.

Pathophysiology of Gout

Prolonged hyperuricemia leads to crystal formation, which may accumulate in synovium and at external sites.

When crystal are phagocytosed there is interaction between the crystals and lysosomal membrane due to weak acid groups and membrane disruption follows.

Phagocytosis of urate leads to increased production of lactic acid which causes further precipitation of crystals.

Clinical features of Gout

- Patient is obese
- Pain usually early morning affecting the big toe.
- Affected joints are usually red, swollen and warm

Precipitating factors of Gout

- Joint trauma
- Unusually physical exercise
- Alcohol
- High protein diet/starvation
- Surgery
- Drugs
- Severe incidental illness

Diagnosis of Gout

- History
- X-ray
- Serum uric level measurements

Management of Gout

The goal of treatment is to:

(i) Reduce acute synovitis
(ii) Prevent further crystallization
(iii) Identify associated disease

Acute synovitis can be managed by NSAIDs and even *Colchicine*.

Management of Gout cont'd

Prevention of crystallization through:

- Weight reduction, especially among obese and overweight individuals
- Reduction of alcohol consumption
- Evaluate need for diuretic drugs

OSTEOPOROSIS

Osteoporosis is a condition characterized by a reduction in bone density and change in bone structure.

In this case the rate of bone resorption is greater than that of bone formation resulting in reduced total bone mass, hence increasing susceptibility to fractures.

Bones are porous, brittle and fragile i.e. fractures easily under stress that would not break a normal bone.

Commonly results in compressed fractures of the thoracic and lumbar spine and intertrochanteric regions of femur.

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Predisposing factors to Osteoporosis

Normal bone remodeling increases bone mass until early 30s.

Gender, race, genetics, aging, low body weight and body mass index, nutrition, lifestyle and physical activity influence peak bone mass and development of osteoporosis.

Osteoporosis is not a disease of the elderly but fractures occur with aging, onset occur in early life when bone mass peak begins to decline.

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Pathophysiology of Osteoporosis

After peak bone mass is achieved, Calcitonin ,which inhibits bone resorption and promotes bone formation is decreased.

Estrogen which inhibits bone breakdown decreases with aging

Parathyroid hormone increases with aging increasing bone turnover and resorption resulting to loss of bone mass

Women develop osteoporosis more than men because of lower bone peak mass and the effect of estrogen loss during menopause

Factors affecting bone mass

- Genetics
- Nutrition
- Age
- Physical exercise
- Lifestyle choices
- Medication

Risk Factors

- Small framed non-obese individuals
- Race
- Sex
- Nutritional factors
- Lifestyle

Diagnosis

- X-Ray
- Laboratory studies (calcium, serum phosphate)

Management of Osteoporosis

(i) ***Adequate nutrition*** rich in calcium and vitamin D throughout life, with an increased calcium intake during adolescence, young adulthood, and the middle years, protects against skeletal demineralization.

(ii) ***Calcium supplementation*** e.g., *Caltrate®, Citrocal®* may be prescribed and taken with meals or with a beverage high in ***Vit. C*** to promote absorption.

(iii) ***Regular weight bearing exercises*** to promotes bone formation.

Management of Osteoporosis cont'd

(iv)Hormonal Replacement Therapy (HRT) with estrogen and progesterone, especially at natural or surgical menopause.

(v)Calcitonin to suppress bone loss.

(vi)If the patient has a fracture, manage appropriately depending on the condition.

Prevention of Osteoporosis

(i) Early identification of at risk teenagers and young adults

(ii) Regular weight bearing exercise.

(iii) Modification of lifestyle (caffeine, cigarettes and alcohol)

(iv) Observe for precipitating factors.

Osteomalacia

Is a metabolic bone disorder characterized by inadequate mineralization of bone.

Faulty mineralization leads to softening and weakening of the skeleton.

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Pathophysiology of Osteomalacia

Deficiency of Vitamin D (calcitriol) which promotes calcium absorption from GI leads to low calcium and phosphate movement to calcification.

Osteo-malacia may result from failed calcium absorption (malabsorption syndrome) or from excessive loss of calcium from the body.

GI disorders –fats are inadequately absorbed are likely to produce osteo-malacia through loss of calcium and Vit D, the first being excreted in feces.

Pathophysiology cont'd

Liver and kidney diseases can produce lack of Vit D because these are organs that convert Vit D to its active form.

The malnutrition type of Osteomalacia is as a result of poverty

Clinical manifestations of Osteomalacia

- Softening and weakening of the skeleton.
- Pain and tenderness to touch.
- Bowing of the bones
- Pathologic fracture
- On exam–skeletal deformity (spinal kyphosis and bowed leg)
- Limping gait

Diagnosis for Osteomalacia

(i) ***History*** of the presenting disorder.

(ii) ***X-ray film*** reveals a generalized demineralization of bone.

(iii) ***Serology*** reveals a low serum calcium and phosphorous level with a moderately elevated concentration of *alkaline phosphatase*.

(iv) ***Urinalysis*** excretion of calcium and creatinine is low.

(v) ***Bone biopsy*** reveals an increase in ***osteoid*** levels.

Management of Osteomalacia

(i) Correct the underlying cause/disorder.

(ii) Increase dose of Vit D with supplement calcium, if the disorder is due to malabsorption. (*High doses of Vit. D are toxic hence adequate monitoring is very necessary).*

(iii) Sunlight exposure to promote Vit. D utilization.

Management of Osteomalacia cont'd

(i) Provide adequate diet/nutrition.

(ii) Use physical, psychological and pharmaceutical measures.

(iii) Long term monitoring of patient progress is required.

BONE TUMOURS

Bone tumors

Neoplasm of the musculoskeletal system are of various types.

They may be ***primary tumors*** or ***metastatic*** from primary cancers .

Metastatic bone tumours are more common than primary bone tumours.

Benign Bone tumours

Benign bone tumours are more common than malignant primary bone tumours.

They are slow growing, well circumscribed, present few symptoms and are not a cause of death.

Types of Benign Bone tumours

(i) ***Osteochondroma***: Is the most common, develops during growth and becomes static. It may become malignant in rare cases, especially, after trauma.

(ii) ***Enchondroma:*** A tumour of hyaline cartilage commonly affecting the hand, femur, tibia and humerus.

Presents with mild ache and pathologic fracture may occur.

Benign Tumours cont.

(iii) Bone cysts

- Osteoid osteoma: A painful tumour that occur in children and young adults.

- Giant Cell Osteoclastoma: Remain benign for a long period but may undergo malignancy transformation, invade and local tissues.

  Are soft, hemorrhagic and common in young adults.

Malignant Bone tumours

Are quite rare and usually arise from connective and supportive tissue cells (sarcomas) or bone marrow elements such as multiple myeloma.

An example is ***Osteosarcoma***: (Osteogenic sarcoma) the most common and often a fatal primary malignant bone tumour.

Prognosis depend on whether it has spread to other organs such as the lungs, liver etc (***Bone tumor metastasis to the lungs is common).***

Appears mostly in males (10 – 25 yrs), older people with ***Paget's disease***, and may also occur following radiation exposure.

Clinical manifestations of Bone tumors

Commonly affects the distal femur, proximal tibia and proximal humerus, presenting with:

(i) Pain

(ii) Swelling

(iii) Limited motion

(iv) Weight loss

(v) Palpable bony mass

Malignant tumour types

(i) Chondrosarcoma - Common among adults.

(ii) Ewing's sarcoma.

(iii) Fibrosarcoma.

(iv) Liposarcoma

Nursing care of a patient with bone tumour

(i) Pain management.

(ii) Evaluation of activities of daily living (ADL).

(iii) Prevention of pathological fracture.

(iv) Promote coping skill.

(v) Promote self esteem.

(vi) Promote understanding of disease process

AMPUTATION

Amputation

Amputation is one of the oldest surgical procedures with artificial limbs identified from over 2000 years ago.

It refers to the removal of the whole or part of an arm/hand or a leg/foot.

Amputations can occur after an injury (traumatic amputation) or deliberately at surgery.

In vascular surgery amputations are only rarely performed on the arms. Vascular surgeons frequently have to perform amputations of toes or legs.

Causes of Amputation

(i) Peripheral vascular disease.

(ii) Gas gangrene.

(iii) Trauma.

(iv) Congenital deformities.

(v) Malignant osteomyelitis/tumour.

Amputations cont'd

A majority of amputations are performed because the arteries of the limbs have been blocked following insufficient blood supply to the limb.

About 30-40% of amputations are performed in patients with diabetes, because diabetes can cause hardening of the arteries. Patients with DM can develop foot/toe ulceration and about 7% of patients will have an active ulcer or a healed ulcer.

Ulcers are recurrent in many patients and approximately 5-15% of diabetic patients with ulcers will ultimately require an amputation.

Types of amputations

Amputations can be divided into **minor** and **major.**

Minor amputations are amputations where only a toe or part of the foot is removed. A ***ray amputation*** is a particular form of minor amputation where a toe and part of the corresponding metatarsal bone is removed.

A forefoot amputation can sometimes be helpful in patients with more than one toe involved by gangrene.

Below knee amputation

This operation can be performed using 2 major techniques. The most common technique is the ***Posterior Myoplastic flap (Burgess) technique*** where the skin and muscle from the calf are brought forward to cover the shin bones after they have been divided.

The other main technique is ***the Skew flap (Kingsley Robinson) technique*** in which the muscles of the calf are brought forward in the same way as in the posterior technique but the skin flaps are skewed in relation to the muscle.

Below knee amputation incision

Above knee amputation

In this procedure the bone in the thigh (femur) is divided about 12-15 cm above the knee joint and the muscle and skin closed over the end of the bone.

Amputation related Complications

(i) Hemorrhage

(ii) Infections

(iii) Skin breakdown

(iv) Phantom limb pain

(v) Joint contractures

Assignment

Read and make notes on the following

(i) Care of the stump.

(ii) Nursing care of patient undergoing amputation.

LOW BACK PAIN

Low Back Pains

Low back pains constitutes one of the most common reasons for seeking medical attention.

Approximately > 84% of adults will have low back pain at some point. Some suffer from chronic or recurrent courses, with substantial impact on quality of life

Most episodes of low back pains are self-limiting

Treatment comprises a wide variety of approaches and no specific approach would be suggested as the most optimum.

Low Back Pains cont'd

Any structure in the back can cause pain, including ligaments, joints, periosteum, musculature, blood vessels, annulus fibrosus and nerves. Intervertebral discs and facet joints being the most commonly affected.

85% of those with isolated low back pains do not have a clear localization. There is no clear histopathology or anatomical location, and is commonly referred to as "***strain***" or "***sprain***".

Age of onset is usually above 35 yrs old with men and women affected equally almost at the same rate.

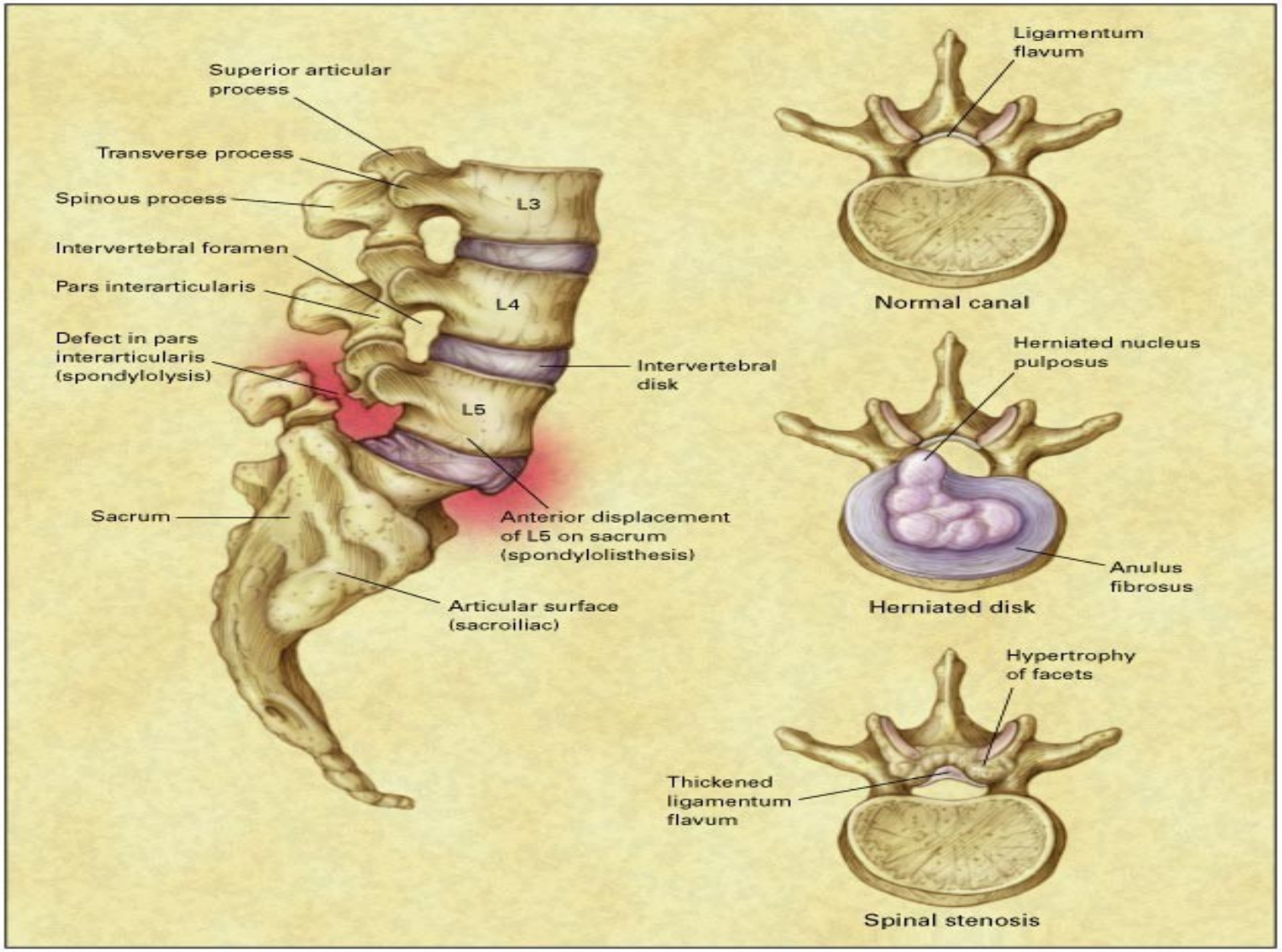
Low Back Pains cont'd

Low back pains is the leading and expensive cause of work disability in those < 45 years

Risk factors Include:

(i) Heavy lifting
(ii) Twisting and vibration exercises
(iii) Obesity.
(iv) Poor conditioning.

mmon Pathoanatomical Conditions of the Lumbar Spine



Diagnosis of Low Back Pain

1. **History**

   Any evidence of systemic disease?

   - Age (especially >50),
   - History of cancer, unexplained weight loss, chronic infection.
   - Duration of illness
   - Presence of nocturnal pain
   - Response to therapy
   - Many patients with infection or malignancy will not have relief when lying down

   For arthritis patients – young age, nocturnal pain and worsening with rest are common

Diagnosis of Low Back Pain cont'd

Any evidence of neurologic compromise e.g.

(i) ***Cauda equina syndrome:*** a medical emergency usually due to a tumor or massive herniation compressing nerves at the *cauda equina*. Urinary retention with overflow, saddle anesthesia, bilateral sciatica, leg weakness, fecal incontinence.

(ii) ***Sciatica:*** a nerve root irritation characterised by sharp/burning pain down the posterior or lateral leg to foot or ankle; can be associated with numbness/tingling. If due to disc herniation often worsens with cough, sneeze or performing the Valsalva.

Diagnosis of Low Back Pain cont'd

Any evidence of neurologic compromise e.g:

(iii) Spinal stenosis: Caused by narrowing of the spinal canal, nerve root canals, or intervertebral foramina.

Most commonly due to bony hypertrophic changes in facet joints and thickening of ***the ligamentum flavum.***

Disc bulging or ***spondylolisthesis*** may also cause back pain. Transient leg tingling, pain in calf and lower extremity that is triggered by ambulation and improved with rest, are common symptoms..

Can be differentiated from ***vascular claudication*** through detection of normal arterial pulses on exam.

Diagnosis of Low Back Pain cont'd

2. Physical Examination

Inspection of back and posture (ie. Scoliosis, kyphosis)

- Range of motion
- Palpation of the spine (vertebral tenderness sensitive for infection)
- If high suspicion of malignancy, do a breast/prostate/lymph node exam
- Peripheral pulses to distinguish from vascular claudication

Physical Examination

Straight leg raise: for those with sciatica or spinal stenosis symptoms

- Patient supine, examiner holds patient's leg straight
- Elevation of less than 60 degrees abnormal and suggests compression or irritation of nerve roots
- Reproduces sciatica symptoms (NOT just hamstring)
- *Ipsilateral straight leg raise* sensitive but not specific for herniated disk
- *Crossed straight leg raise* (symptoms of sciatica reproduced when opposite leg is raised) insensitive but highly specific

Physical examination

- Neurologic examination
  - L5: ankle and great toe dorsiflexion
  - S1: plantar flexion, ankle reflex

- Dermatomal sensory loss
  - L5: numbness medial foot and web space between 1st and 2nd toes
  - S1: lateral foot/ankle

Imaging

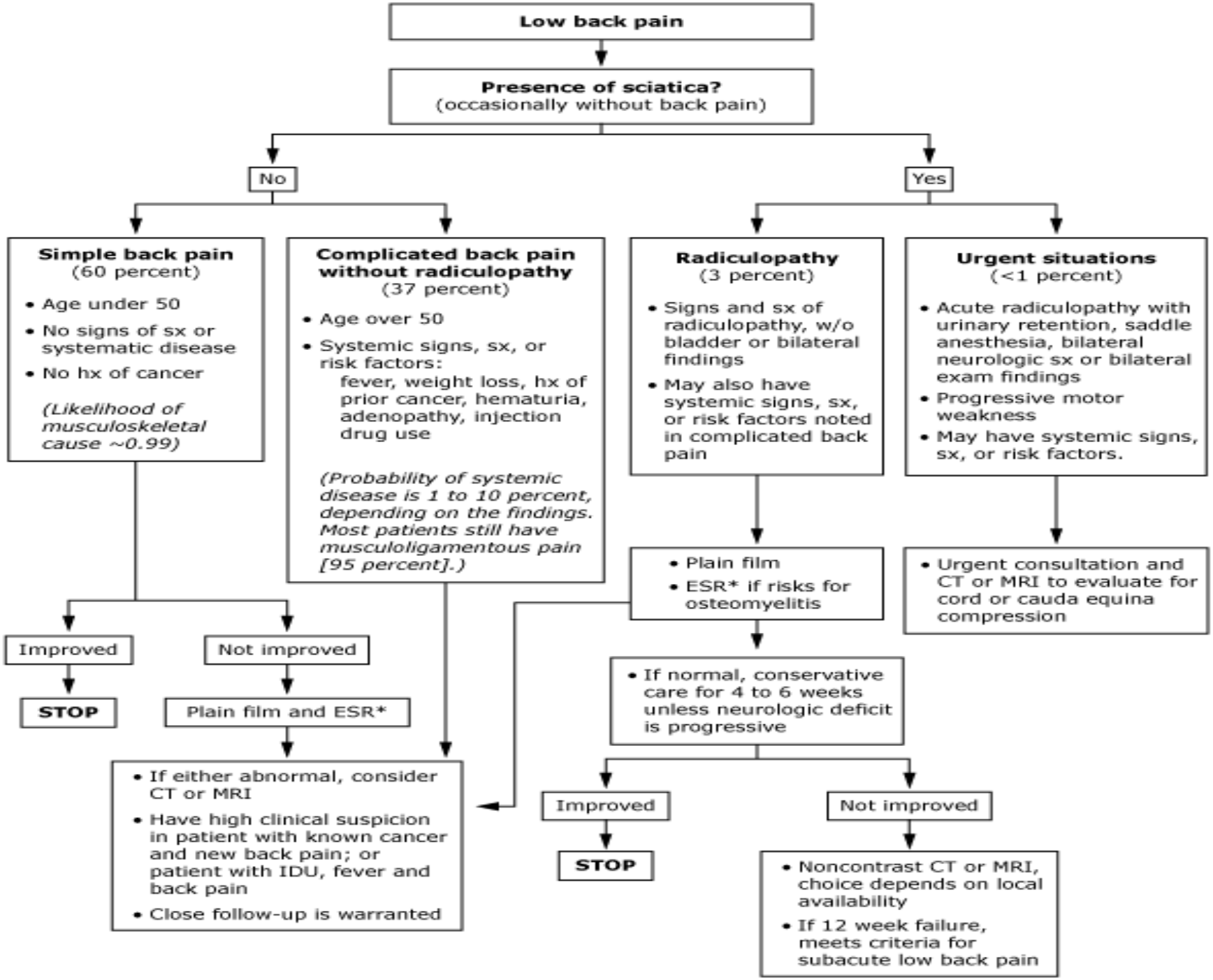
- AP and lateral L-spine if no clinical improvement after 4-6 weeks

"Clinicians should not routinely obtain imaging or other diagnostic tests in patients with nonspecific low back pain" (American College of Physicians and American Pain Society).

  - Do perform x-rays if: fever, unexplained weight loss, hx of cancer, neurologic deficits, EtOH, age <18 or >50, trauma, immunosuppression, prolonged steroid use, skin/urinary infection, indwelling catheter

Imaging

- CT and MRI
  - More sensitive for detection of infection and cancer than plain films
  - Also able to image herniated discs and spinal stenosis, which cannot be appreciated on plain films
  - Beware: herniated/bulging discs often found in asymptomatic volunteers may lead to overdiagnosis/overtreatment
  - MRI better than CT for detection of infection, metastases, rare neural tumours


Low back pain
Presence of sciatica?
(occasionally without back pain)
No
Yes
Simple back pain
(60 percent)
• Age under 50
• No signs of sx or systematic disease
• No hx of cancer
(Likelihood of musculoskeletal cause ~0.99)
Complicated back pain without radiculopathy
(37 percent)
• Age over 50
• Systemic signs, sx, or risk factors:
fever, weight loss, hx of prior cancer, hematuria, adenopathy, injection drug use
(Probability of systemic disease is 1 to 10 percent, depending on the findings. Most patients still have musculoligamentous pain [95 percent].)
Radiculopathy
(3 percent)
• Signs and sx of radiculopathy, w/o bladder or bilateral findings
• May also have systemic signs, sx, or risk factors noted in complicated back pain
Urgent situations
(<1 percent)
• Acute radiculopathy with urinary retention, saddle anesthesia, bilateral neurologic sx or bilateral exam findings
• Progressive motor weakness
• May have systemic signs, sx, or risk factors.
Improved
STOP
Not improved
Plain film and ESR*
• If either abnormal, consider CT or MRI
• Have high clinical suspicion in patient with known cancer and new back pain; or patient with IDU, fever and back pain
• Close follow-up is warranted
• Plain film
• ESR* if risks for osteomyelitis
• If normal, conservative care for 4 to 6 weeks unless neurologic deficit is progressive
Improved
STOP
Not improved
• Noncontrast CT or MRI, choice depends on local availability
• If 12 week failure, meets criteria for subacute low back pain
• Urgent consultation and CT or MRI to evaluate for cord or cauda equina compression

Natural History

- Most recover rapidly
  - 90% of patients seen within 3 days of symptom onset recovered within 2 weeks
- Recurrences are common
  - Most have chronic disease with intermittent exacerbations
- Spinal stenosis is the exception usually gets progressively worse with time

Management of Low Back Pain

- Non-specific low back pain
  - Regular NSAIDs and muscle relaxants good for symptomatic relief.
  - Spinal manipulation of limited utility in studies
  - Should recommend rapid return to normal activities with neither bed rest nor exercise in the acute period
    - Bed rest found to not improve and may delay recovery
  - Exercises not useful in acute phase; use in chronic

Therapy

- Nonspecific low back pain
  - Traction, facet joint injections have minimal effects.
  - Systematic reviews of ***acupunture*** have shown little benefit
  - ? Massage therapy have some promising results
  - Surgery only effective for sciatica, spinal stenosis or spondylolisthesis

Therapy

- Herniated intervertebral discs
  - Nonsurgical treatment for at least a month
    - Exceptions: cauda equina syndrome, progressive neurologic deficits
  - Early treatment same as for nonspecific low back pain, but may need short courses of narcotics for pain control
  - Bed rest not useful
  - Some patients benefit from epidural corticosteroid injections
  - If severe pain, neurologic defecits consider MRI and surgery.

Therapy

- Spinal stenosis
  - Physiotherapy to reduce risk of falls
  - Analgesics, NSAIDs, epidural corticosteroids
  - Decompressive laminecotomy
  - Spinal fusion with decompression if there is additional spondylolisthesis
  - Symptoms often recur, even after successful surgery

Therapy

- Chronic low back pain
  - Intensive exercise improves function and reduces pain, but is difficult to adhere to
  - Anti-depressants: many with chronic low back pain are also depressed
    - ? Maybe for those without depression (tricyclics)
  - Opiates
    - Small RCT showed better effect on pain and mood than NSAIDs
    - No improvement in activity
    - Significant side effects: drowsiness, constipation, nausea

Therapy

- Chronic low back pain
  - Referral to multidisciplinary pain center
    - Cognitive-behavioural therapy, education, exercise, selective nerve blocks
  - Surgical procedures rarely helpful

SPONDYLOARTHRITIS

Introduction

- Spondyloarthritis
  - Refers to inflammatory changes involving the spine and the spinal joints.
    - Remember – can sometimes have peripheral arthritis without spinal symptoms!
- Seronegative Spondyloarthritis
  - Absence of Rheumatoid Factor
    - Psoriatic Arthritis
    - Ankylosing Spondylitis
    - Reactive Arthritis
    - Enteropathic Arthritis
    - Undifferentiated Spondyloarthropathy

Inflammatory vs. Mechanical Back Pain

- ***Inflammatory***
  - Age of onset < 40
  - Insidious onset
  - > 3 months duration
  - > 60 min am stiffness
  - Nocturnal pain
  - Improves with activity
  - Tenderness over SI joints
  - Loss of mobility in all planes
  - Decreased chest expansion
  - Unlikely to have neurologic deficits

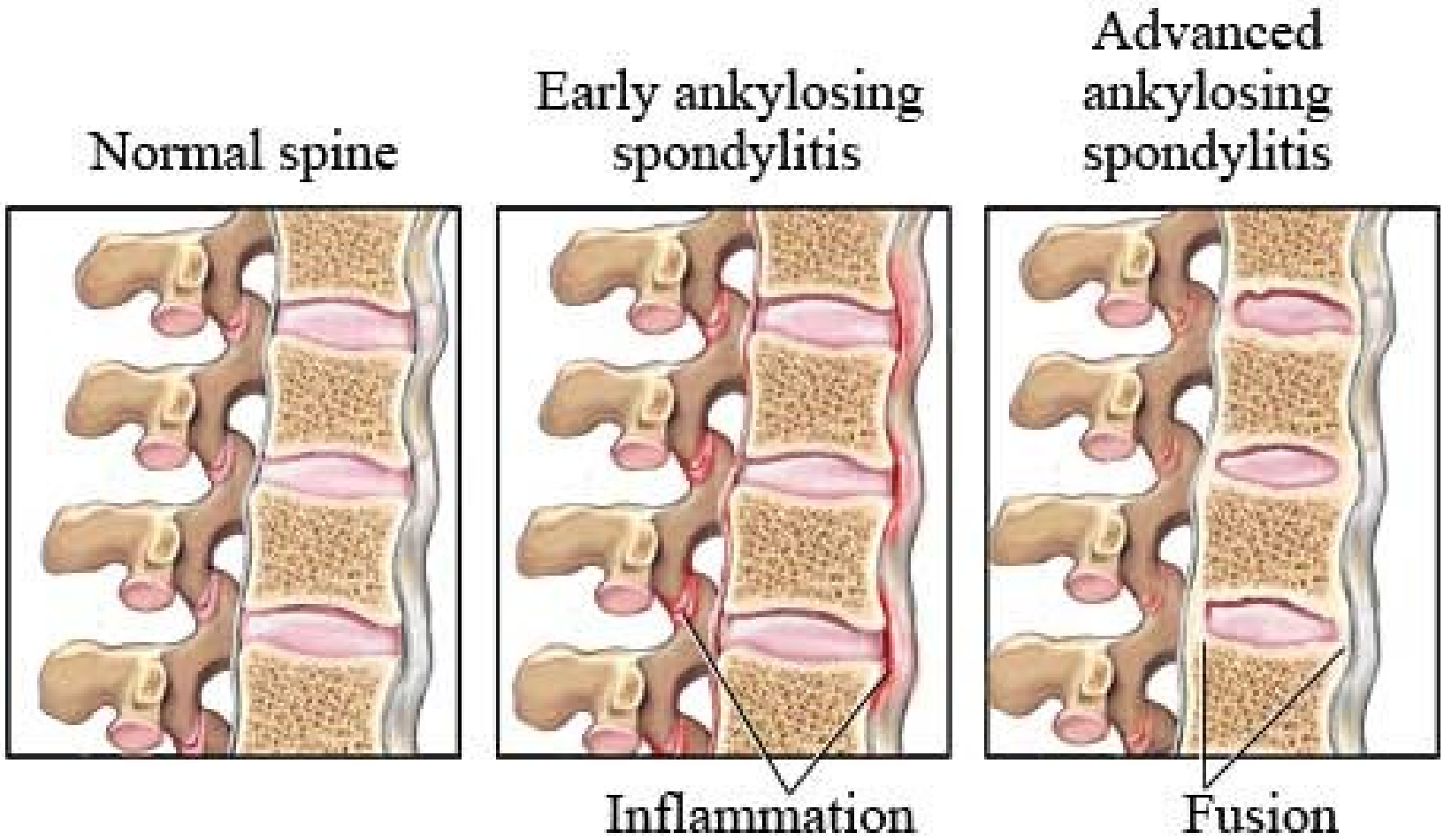
- ***Mechanical***
  - **Any age**
  - **Acute onset**
  - **< 4 weeks duration**
  - **< 30 min am stiffness**
  - **No nocturnal pain**
  - **Worse with activity**
  - **No SI joint tenderness**
  - **Abnormal flexion**
  - **Normal chest expansion**
  - **Possible neurologic deficits**

Clinical Features

Sacroiliitis

- Usually bilateral and symmetric
- Initially involves the synovial-lined lower 2/3 of the Sacro-iliac joint.
- Early changes involves erosion on the iliac side of SI joint , the cartilage becomes thinner, and can cause “pseudo-widening” of SI joint.
- Bony sclerosis, then complete bony

Spinal Involvement



Spinal Involvement

- Gradual ossification of the outer layers of the annulus fibrosis (***Sharpey's fibers***) form interverterbral bony bridges called ***syndesmophytes***.

- Fusion of the apophyseal joints and calcification of the spinal ligaments along with bilateral syndesmophyte formation can result in "***bamboo spine***"

Enthesitis

- ***Enthesis***: site of insertion of ligament, tendon or articular capsule into bone.

- ***Enthesitis***: Is the inflammation of enthesis resulting in new bone formation or fibrosis.

- Common sites: SI joints, intervertebral discs, manubriosternal joints, symphysis pubis, iliac crests, trochanters, patellae, clavicles, calcanei (Achille's or plantar fasciitis)

Treatment of Enthesitis

1. ***NSAIDs***
   - Good for mild symptoms
   - Potentially disease modifying
   - Indomethacin seems to work the best
   - Beware of side effects, especially gastrointestinal disease

Treatment of Enthesitis

2. ***Disease – Modifying Anti-Rheumatic Drugs (DMARDs)***.

- Sulfasalazine 1000-2000 mg bid

  Seems to be the most effective for spinal symptoms.

- Methotrexate 15-25 mg weekly
  - For patients with prominent peripheral arthritis
  - Doesn’t work very well for spinal symptoms

Treatment

3. *Steroids*

- Not very effective at all in AS
- Local injections for enthesitis or peripheral arthritis

4. *Anti-TNFα agents*

- Remicade (infliximab), Enbrel (etanercept) and Humira (adalimumab)
  - Very useful for treating symptoms, improving ROM, improving fatigue
  - Hopefully disease-modifying . . .

MUSCULOSKELETAL DEFORMITIES

Musculoskeletal deformities

1.Congenital dislocation of the hip

- Unilateral or bilateral hip dislocation can be diagnosed soon after.

- If the diagnosis is missed the child will develop shortening and external rotation of the affected limb.

CLUB FOOT

Kids at cure international

CLUB FOOT

THE NORMAL FOOT

Complex organ that is required to be

- **Stable:** for supporting the body weight in standing
- **Resilient:** for walking and running
- **Mobile:** to accommodate variations of surface
- **Cosmetic**

CLUB FOOT

Definitions

Talipes: **Talus = ankle**
Pes = foot

Equinus: **(Latin = horse)**
Foot that is in a position of planter flexion at the ankle, looks like that of the horse.

Calcaneus: **Full dorsiflexion at the ankle**

CLUB FOOT

Planus: **flatfoot**

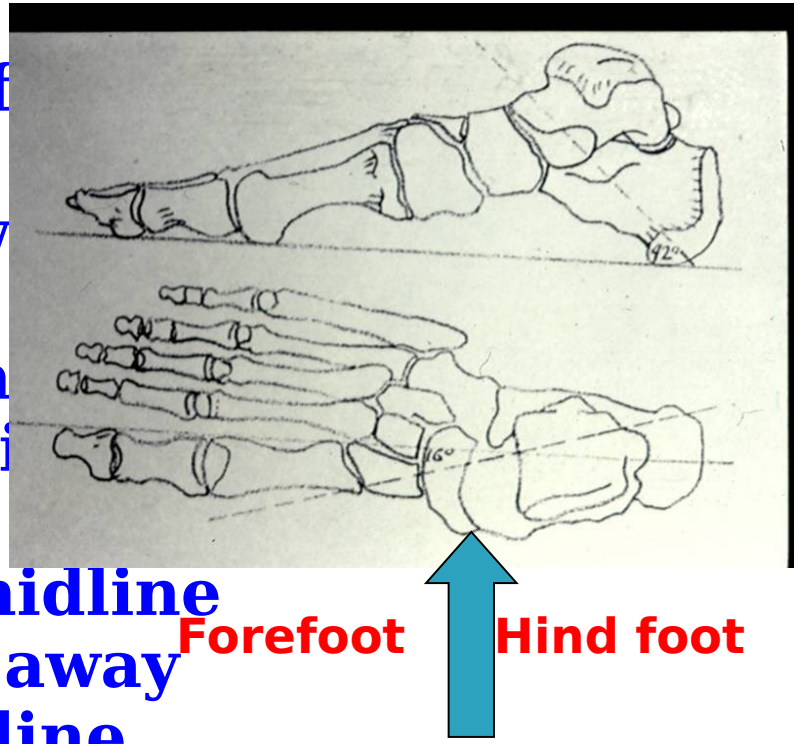
Cavus: **highly arched f**

Varus: **heal going tow the midline**

Valgus: **heel going awa from the midli**

Adduction: **forefoot going towards the midline**

Abduction: **forefoot going away From the midline**



CLUB FOOT

Types

Idiopathic (Unknown Etiology) :

- **Congenital Talipes Equino-Varus CTEV**

Acquired, Secondary to :

- **CNS Disease : Spina bifida, Poliomyelitis**
- **Arthrogryposis**
- **Absent Bone : fibula / tibia**

Congenital Talipes Equino-Varus CTEV

Etiology

Some of these factors are :

- Abnormal intrauterine forces
- Arrested fetal development
- Abnormal muscle and tendon insertions
- Abnormal rotation of the talus in the mortise

Congenital Talipes Equino-Varus CTEV

Incidence

- Occurs approximately in one of every 1000 live birth
- In affected families, clubfeet are about 30 times more frequent in offspring
- Male are affected in about 65% of cases
- Bilateral cases are as high as 30 – 40 %

Congenital Talipes Equino-Varus CTEV

Basic Pathology

- **Abnormal Tarsal Relation**
  Congenital Dislocation / Subluxation

- **Soft Tissue Contracture**
  Congenital Atresia

Congenital Talipes Equino-Varus CTEV

Diagnosis

General Examination :

Exclude

- Neurological lesion that can cause the deformity "Spina Bifida"
- Other abnormalities that can explain the deformity
- Presence of concomitant congenital anomalies

- Syndromatic clubfoot

Congenital Talipes Equino-Varus CTEV

Diagnosis

“ Hind foot “
Equinus, Varus

“ Fore foot “
Adduction, Supination, Cavus

Congenital Talipes Equino-Varus CTEV

Diagnosis

- Short Achilles tendon
- High and small heel
- No creases behind Heel
- Abnormal crease in middle of the foot
- Foot is smaller in unilateral affection
- Callosities at abnormal pressure areas
- Internal torsion of the leg
- Calf muscles wasting
- Deformities don’t prevent walking

Congenital Talipes Equino-Varus CTEV

Treatment

The goal of treatment for clubfoot is to obtain a plantigrade foot that is functional, painless, and stable over time

A cosmetically pleasing appearance is also an important goal sought by the surgeon and the family

Congenital Talipes Equino-Varus CTEV

Treatment

Non surgical treatment should begin shortly after birth

1. Gentle manipulation

2. Immobilization
   - Strapping ????
   - POP or synthetic cast

Congenital Talipes Equino-Varus CTEV

Treatment

Non surgical treatment should begin shortly after birth

3. Splints to maintain correction
   - Ankle-foot orthosis ????
   - Dennis Brown splint

Congenital Talipes Equino-Varus CTEV

Treatment

Manipulation and serial casts

- Validity, up to 6 months !
- Technique “Ponseti”
- Avoid false correction
- When to stop ?
- Maintaining the correction
- Follow up to watch and avoid recurrence

Congenital Talipes Equino-Varus CTEV

Treatment

Ponseti technique

1. Always use long leg casts, change weekly.
2. First manipulation raises the 1st metatarsal to decrease the cavus
3. All subsequent manipulations include pure abduction of forefoot with counter-pressure on neck of talus.
4. Never pronate !
5. Never put counter pressure on calcaneus or cuboid.

Congenital Talipes Equino-Varus CTEV

Treatment

Ponseti technique (cont.)

6. Cast until there is about 60 degrees of external rotation (about 4-6 casts)
7. Percutaneous tendo Achilles tenotomy in cast room under local anesthesia, followed by final cast (3 weeks)
8. After final cast removal, apply Normal last shoes with Denis Browne bar set at 70 degrees external rotation (40 degrees on normal side)
9. Denis Browne splint full time for two months, then night time only for two-four years.
10. 35% need Anterior Tibialis tendon transfer at age 2-3

Congenital Talipes Equino-Varus CTEV

Surgical Treatment

Indications

- Late presentation, after 6 months of age !
- Complementary to conservative treatment
- Failure of conservative treatment
- Residual deformities after conservative treatment
- Recurrence after conservative treatment

Congenital Talipes Equino-Varus CTEV

Surgical Treatment

- Types (soft tissue and bony operations)
- Time of surgery
- Selection of the procedure and the incision
- Post operative care
- Follow up
- Complications

Congenital Talipes Equino-Varus CTEV

Surgical Treatment

Soft tissue operations

1. Release of contractures
2. Tenotomy
3. Tendon elongation
4. Tendon transfer
5. Restoration of normal bony relationship

Congenital Talipes Equino-Varus CTEV

Surgical Treatment

Bony operations

- Indications
- Usually accompanied with soft tissue operation
- Types:

  - Osteotomy, to correct foot deformity or int. tibial torsion
  - Wedge excision
  - Arthrodesis (usually after bone maturity) one or several joints
  - Salvage operation to restore shape.

Congenital Talipes Equino-Varus CTEV

Surgical Treatment

Congenital Talipes Equino-Varus CTEV

Surgical Treatment